A Clustering Algorithm Based on Dual-Perspective Adaptive Conical Information Granules

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***Abstract*—Traditional clustering algorithms often rely on fixed distance metrics and single-feature perspectives, making them sensitive to parameter settings and noise, particularly on datasets with irregular shapes or uneven density. Inspired by granular computing theory, we propose a dual-perspective adaptive conical information granule clustering algorithm (ADGC). ADGC integrates a dynamic threshold mechanism with an adaptive granulation strategy. First, a data-driven dominance threshold is established using attribute quantile entropy. Then, upper and lower conical information granules are constructed from two feature perspectives, with radii dynamically estimated based on distance growth trends to effectively filter noise. To ensure robust convergence, initial cluster centers are selected using decision values derived from global bidirectional granule centrality and spatial dispersion constraints. For each bidirectional granule, coverage strength and weight are computed to generate co-occurrence similarity matrices, which are merged across perspectives to overcome single-view limitations. A center-weighted mechanism iteratively updates cluster assignments, enhancing data association, adaptability, and generalization on heterogeneous datasets. Comparative experiments demonstrate that ADGC accurately identifies clusters and outperforms mainstream granular computing-based algorithms on complex datasets with overlapping clusters.**

***Index Terms*—Information granule, bidirectional granular structure, coverage strength, clustering.**

I. Introduction

Cluster analysis is an unsupervised learning approach that groups datasets into clusters by measuring similarity between data objects. This ensures that intra-cluster samples exhibit high similarity, while inter-cluster differences remain significant, thus enabling effective structural analysis of the data [1]. Due to its independence from labeled data, clustering has found widespread applications in machine learning, data mining, and bioinformatics [2].

Over the past few decades, a wide range of clustering algorithms have been developed. These can generally be categorized into four main types: hierarchical, partition-based, density-based, and graph-based algorithms [3]. While these classical methods have demonstrated effectiveness in many scenarios, they struggle to scale with the rising complexity of modern data, which is often high-dimensional, non-Euclidean, and heterogeneously distributed [4].

For instance, partitioning methods such as K-means++ rely on Euclidean distances and are prone to local optima, particularly when handling irregular cluster shapes or noisy data [5]. Hierarchical algorithms are sensitive to outliers and cannot reverse early-stage erroneous splits or merges. Density-based methods (e.g., DBSCAN) depend heavily on fixed density thresholds, making them less effective for multi-density datasets. Graph-based approaches often suffer from prohibitive computational costs under complex or large-scale data.

In response to these limitations, granular computing has emerged as a promising paradigm for enhancing clustering algorithms. It decomposes complex data into multi-granular information structures, enabling noise reduction and semantic preservation through hierarchical abstraction. This multi-level representation facilitates the identification of structural patterns that traditional methods often overlook.

Recent efforts have applied granular computing principles to classic clustering frameworks. For example, Xia et al. [6] embedded granular balls into k-means to reduce distance computations. Yu et al. [7] introduced a hierarchical method leveraging attribute-based granulation to improve clustering on non-Euclidean data. Ouyang et al. [8] constructed flexible granules for DBSCAN to remove shape constraints. Chen et al. [9] enhanced online clustering for uncertain data via a mean-shift algorithm based on granule neighborhoods. Hu et al. [10] extended fuzzy clustering by embedding triangular knowledge granules into the FCM algorithm, improving adaptability to numerical uncertainty.

Several granular ball-based algorithms have also been proposed to improve scalability and robustness. Xia et al. [11] presented GB-DBSCAN to reduce computational overhead via unsupervised granular ball generation. Wang et al. [12] proposed GB-DP, addressing Density Peak clustering's limitations by modeling data distribution using granules. Tang et al. [10] tackled information uncertainty using fuzzy granules to mitigate clustering inaccuracies and time costs.

To further enhance robustness and expressiveness, coarse-grained representations have been adopted. Xia et al. [13] introduced the GBC algorithm, using granular balls and inter-ball relations to achieve robust clustering. Jin et al. [14] developed a multi-objective version of GBC, optimizing alignment loss by balancing intra-cluster compactness and inter-cluster separation. Zhang et al. [15] introduced a justified granularity generation

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method (GB-POJG) using coverage and specificity for quality evaluation and anomaly detection. Xie et al. [16] proposed a granular-ball-based spectral clustering method to reduce computational complexity. In contrastive learning, Su et al. [17] incorporated granular balls for multi-view clustering, preserving both intra- and inter-view structure. Cheng et al. [18] applied granular anchors in manifold learning to construct efficient bipartite graphs for low-dimensional embeddings.

These advances demonstrate that granule-based clustering techniques provide significant improvements in terms of efficiency, adaptability, and structure preservation. However, existing methods often rely on single-perspective granulation, fixed thresholds, or complex quality metrics. Such dependencies introduce sensitivity to parameter settings and hinder generalization when applied to data with diverse structures or overlapping distributions.

To address these limitations, we propose a novel dual-perspective Adaptive Conical Granule Clustering (ADGC) algorithm. This approach dynamically constructs upper and lower conical information granules to model bidirectional dominance relationships in the data. It evaluates granule centrality, selects representative cluster centers via similarity and distance filtering, and integrates dual-perspective similarity matrices to enhance structural association. Furthermore, an iterative, center-guided assignment strategy refines cluster structures, enabling robust clustering for data with varying density and distribution.

The key contributions of this paper are summarized as follows:

- We propose a granule modeling framework that dynamically generates dominance thresholds using quantile entropy and adaptively estimates neighborhood radii based on granule compactness trends. This enables flexible construction of global and local conical granules, achieving noise-resilient modeling across diverse data types.
- We introduce a granule optimization strategy for cluster center selection, leveraging both positively and negatively dominated neighborhoods along with relative spatial positioning. This enhances the representativeness and information richness of selected centers.
- We design a novel dual-perspective similarity matrix based on granule coverage strength and weight, capturing local associations and mitigating single-view limitations. This improves performance in low-density and heterogeneous data scenarios.

The remainder of the paper is organized as follows: Section II reviews related work on clustering and conical information granules. Section III presents the proposed ADGC algorithm. Section IV reports the experimental results.

II. Preliminaries

The proposed algorithm in this paper builds upon the center-based partitioning framework, while incorporating the hierarchical abstraction concept from granular computing. Unlike conventional approaches that rely solely on Euclidean distances or local density for defining sample associations, our method introduces a bidirectional conical information granule structure as a global representation mechanism for each data object. This structure constructs a dual-perspective semantic system, capturing both global and local dominance-subordination relationships, through the object ordinal relation matrix. This transition from conventional distance-based clustering to a semantics-driven representation based on structural associations offers a novel perspective for exploring latent cluster structures in complex data. This section provides the necessary theoretical background by reviewing center-based clustering approaches, the concept of information entropy, and the construction of bidirectional conical granules, laying the foundation for the adaptive similarity computation introduced in later sections.

A. Center-Based Clustering

Center-based clustering is one of the most fundamental paradigms in machine learning and data mining. Its core idea is to select representative cluster centers and assign data points to their nearest centers, forming compact intra-cluster structures and well-separated inter-cluster regions. The quality of the final clustering outcome heavily depends on the effectiveness of center selection.

This concept was first introduced by the K-means algorithm proposed by MacQueen [19], which minimizes the sum of squared distances between samples and their closest cluster centers through an iterative partition-update process. However, K-means is sensitive to initial center selection and can easily converge to local optima. To mitigate this issue, K-means++ [20] incorporated a distance-based initialization strategy, selecting initial centers that are far apart to improve convergence speed and result stability. To handle nonlinear distributions, Kernel K-means [21] maps data into a high-dimensional feature space using kernel functions. While this allows better modeling of complex cluster shapes, it introduces higher computational costs and challenges in kernel parameter selection. Mean-Shift clustering [22] identifies cluster centers as local maxima of a probability density function estimated via kernel density estimation. This approach can discover arbitrarily shaped clusters without specifying the number of clusters in advance, but it is highly sensitive to bandwidth parameters and computationally intensive. Density Peak Clustering (DPC) [23] employs a dual criterion of high density and large distance to identify cluster centers, improving robustness to noise and non-convex shapes. Nevertheless, it struggles to distinguish regions with smoothly varying densities. Recent works have proposed enhanced center identification mechanisms. Chen et al. [24] introduced a fast density-guided center selection strategy to accelerate initial center detection. Attila et al. [25] defined cluster gravity centers using a critical distance threshold, enhancing adaptability to arbitrarily shaped clusters. Xu et al. [26] incorporated a local update mechanism into the K-means framework to reduce initialization randomness and improve stability. Wang et al. [27] proposed a variational DPC method that first identifies representative points and then performs hierarchical refinement, integrating DPC and DBSCAN principles for robust center detection under arbitrary density distributions.

Despite these advances, many methods still rely on Euclidean distance and density estimation, facing two major limitations:

(1) Euclidean distance may fail to capture similarities in complex or non-convex distributions, reducing cluster representativeness; (2) these models are often sensitive to initialization and parameter settings, leading to unstable or non-reproducible results. These challenges motivate the development of adaptive, granular computing-inspired center-based clustering approaches.

B. Information Entropy

Information entropy is a fundamental concept in information theory and a widely used tool in modern information processing for quantifying uncertainty. By computing entropy, one can evaluate the variability of data distributions and measure differences in information content across distinct states. In clustering, entropy serves as a robust criterion for identifying informative attributes and reducing intra-cluster ambiguity, thereby supporting effective pattern discovery in large and complex datasets [28].

Formally, consider a discrete random variable X with a value set $\{x_1, x_2, \ldots, x_n\}$ and corresponding probability distribution $\{p_1, p_2, \ldots, p_n\}$, where $\sum_{i=1}^{n} p_i = 1$. The entropy of X, denoted as $H(X)$, is defined as:

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i \tag{1}$$

A higher entropy indicates greater uncertainty or disorder in the variable. Conceptually, entropy represents the average self-information of all possible outcomes and provides a foundation for multivariate information analysis.

C. Bidirectional Conical Information Granule

To reduce information loss in clustering, the concept of bidirectional conical information granules (BCIG) has been proposed within the granular computing framework [29]. BCIGs granularize objects based on ordinal relationships across all dimensions, leveraging the semantic information embedded in conical granules to capture object characteristics more comprehensively. Using the majority of attribute conditions as the granulation criterion, the method analyzes correlations among object neighborhoods to construct BCIGs for all objects.

Formally, an information system (IS) is defined as a quadruple $S = \{U, A, f, V\}$, where $U = \{x_1, \ldots, x_n\}$ is a finite set of objects, $A = \{a_1, \ldots, a_m\}$ is a set of attributes, V_a denotes the value domain of attribute a, $V = \bigcup_{a \in A} V_a$, and $f : U \times A \to V$ is an information function. Each object $x_i \in U$ is represented as an m-dimensional vector $(f(x_i, a_1), \ldots, f(x_i, a_m))$, and $d(x_i, x_j)$ denotes the Euclidean distance between objects.

Given S and an object $x_i \in U$, the upper conical granule (UG) of x_i is defined as:

$$\begin{aligned} EC_B^{\Uparrow}(x_i) = \{&y \in U \mid \exists B \subseteq A \text{ s.t.} \\ &\times (\forall a \in B, f(y, a) \leq f(x_i, a))\} \end{aligned} \tag{2}$$

The lower conical granule (LG) is defined as:

$$\begin{aligned} IC_B^{\Downarrow}(x_i) = \{&y \in U \mid \exists B \subseteq A \text{ s.t.} \\ &\times (\forall a \in B, f(y, a) \geq f(x_i, a))\} \end{aligned} \tag{3}$$

where $|B| \geq \lceil \frac{|A|}{2} \rceil$.

Here, $EC_B^{\Uparrow}(x_i)$ represents the advantageous ordinal conical granule of object x_i, while $IC_B^{\Downarrow}(x_i)$ represents the disadvantageous granule. These dual granules jointly explore object neighborhoods from complementary ordinal perspectives, reducing uncertainty and enriching the discovery of cluster structures.

The UG captures samples that do not exceed the object in most attribute dimensions, reflecting forward coverage in the structural space. Conversely, the LG describes samples that are not lower than the object, representing reverse structural constraints. UG emphasizes the object's ability to include neighboring samples, whereas LG highlights samples that structurally dominate it. By jointly modeling coverage and constraint relations, the dual-view formulation mitigates information loss inherent in single-view representations based solely on local distances or one-sided ordering.

To efficiently represent relationships among granules for all objects, two binary matrices are defined:

$$\mathrm{DB}^{\Uparrow}(x_i, x_j) = \begin{cases} 1, & x_j \in EC_B^{\Uparrow}(x_i) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\mathrm{IDB}^{\Downarrow}(x_i, x_j) = \begin{cases} 1, & x_j \in IC_B^{\Downarrow}(x_i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

These matrices capture neighborhood semantics and preserve self-referential granularity, ensuring that each object is included in its own granule. The semantic structure provided by conical granules forms a richer representation of objects, serving as the foundation for subsequent similarity computation and clustering optimization.

III. Algorithm

In this section, we provide a detailed description of the Dual-Perspective Adaptive Conical Information Granule Clustering (ADGC) algorithm. We first introduce the threshold selection strategy employed in the construction of conical information granules. Next, we describe the formulation of adaptive R-upper and R-lower conical granules, along with the corresponding similarity measures derived from dual perspectives. Building upon these foundations, we present the complete clustering procedure, which includes cluster center selection, partition-based assignment of objects, and the integration of clustering results from the two semantic perspectives. This structured exposition highlights the key innovations of ADGC and prepares the reader for the subsequent algorithmic details.

A. Threshold Selection

In the construction of conical granules, the threshold θ controls granule inclusion and directly affects granule sparsity. To accommodate diverse data distributions and heterogeneous attribute complexities, we determine θ using an entropy-oriented rule based on quantile entropy (QE). QE quantifies attribute-wise uncertainty, with higher dispersion indicating the need for a broader inclusion threshold. By reflecting dataset-specific variability, this approach dynamically adapts the threshold θ, enhancing the expressiveness and robustness of the resulting granule structures.

Definition 3.1: Let $S = \{U, A, f, V\}$ be an IS, where $A = \{a_1, \ldots, a_m\}$ is the set of m attributes. For each attribute a_i, its values are divided into q equal-frequency quantile intervals, with the proportion of samples in the j-th interval denoted as p_j. The quantile entropy (QE) of a_i is defined as:

$$QE(a_i) = -\sum_{j=1}^{q} p_j \log(p_j) \tag{6}$$

The adoption of QE for threshold selection in the ADGC framework stems from the ordinal nature of dominance relations underlying conical granules. In this setting, clustering decisions are driven by rank consistency rather than precise numerical distances. Unlike conventional dispersion measures such as variance (VAR), interquartile range (IQR), and median absolute deviation (MAD), which operate in value space and quantify numerical spread, QE measures uncertainty in rank space by modeling how samples distribute across quantile partitions. Because it relies on ordinal relationships rather than absolute attribute values, QE provides a distribution-aware description of dispersion that is inherently robust to scale and value magnitude. This property is particularly important for dominance-based clustering. The construction of conical granules depends on ordinal comparisons among objects; therefore, threshold selection should reflect uncertainty in ranking rather than variability in raw values. By capturing the full distribution over quantile intervals, QE offers a stable and adaptive criterion for threshold determination, which directly influences the formation of conical granules and, consequently, the resulting clustering structure. Hence, QE is naturally aligned with the dominance-based semantics of the ADGC algorithm and provides a more principled measure of ordinal uncertainty than traditional value-space dispersion metrics.

The entropy values are normalized to form a relative entropy vector $\hat{QE}_i$, where higher values indicate greater attribute dispersion. To capture the overall complexity of attribute distributions, we define a quantile-based parameter $\tau \in (0.5, 1)$ and compute the proportion of high-complexity attributes:

$$P = \frac{|\{i \mid \hat{QE}_i > \text{quantile}(\hat{QE}, \tau)\}|}{m}, \tag{7}$$

where quantile$(\hat{QE}, \tau)$ denotes the τ-quantile of the normalized entropy vector.

To prevent overly dense connections (from too low thresholds) or excessively sparse granules (from too high thresholds), a piecewise function is used to adaptively bound θ:

$$\theta = \begin{cases} 0.5 + \frac{1-P}{2}, & \text{if } P > 0.5, \\ 0.5 + \frac{P}{2}, & \text{otherwise.} \end{cases} \tag{8}$$

Here, 0.5 serves as a neutral boundary, distinguishing majority support (> 0.5) from non-majority support (≤ 0.5), in line with the majority-agreement principle underlying conical granules.

By adjusting granule inclusion thresholds according to entropy-informed complexity, this method ensures flexible and precise granule formation, particularly for datasets with heterogeneous or overlapping structures.

B. Adaptive R-Upper and R-Lower Conical Information Granules

Traditional conical granules rely on global ordinal dominance to define neighborhoods, which may include irrelevant or redundant neighbors, particularly in heterogeneous or noisy datasets. This can degrade model precision and increase computational cost. To address this, we incorporate an *adaptive neighborhood truncation mechanism* for both the upper (UG) and lower (LG) conical granules. Instead of fixed-radius criteria, the neighborhood boundary is dynamically adjusted based on structural compactness trends, maintaining compatibility with the original conical granule framework while enhancing robustness.

1) Adaptive Radius Mechanism: The neighborhood expansion begins at the cone apex (i.e., the target object) and iteratively includes the nearest unvisited points within the conical structure. To quantify structural consistency during expansion, we define the compactness of a newly added object x_j relative to the current neighborhood $\mathcal{S}_i$ as:

Definition 3.2: Let $EC_B^{\Uparrow}(x_i)$ denote the upper granule of object x_i, and $\mathcal{S}_i = \{x_1, x_2, \ldots, x_k\}$ the currently expanded neighborhood. The compactness of $x_j \in EC_B^{\Uparrow}(x_i)$ with respect to $\mathcal{S}_i$ is defined as:

$$C^{(t)} = \frac{1}{|\mathcal{S}_i|} \sum_{x_k \in \mathcal{S}_i} d(x_j, x_k), \tag{9}$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance.

By recording $C^{(t)}$ after each inclusion, we obtain the compactness sequence $\{C^{(t)}\}_{t=1}^{\infty}$, capturing the evolving local structure. The relative growth rate of compactness is then computed as:

$$r^{(t)} = \frac{C^{(t)} - C^{(t-1)}}{C^{(t-1)}}. \tag{10}$$

During neighborhood expansion, objects within the same cluster lead to slow increases in compactness, whereas inclusion of boundary or noise points causes a sharp rise in $r^{(t)}$. Consequently, consecutive large increases in $r^{(t)}$ indicate that the neighborhood has transitioned from a dense, consistent structure to a dispersed, unstable region, signaling a suitable truncation point.

Let $\mathcal{R}^{(t)} = \{r^{(1)}, r^{(2)}, \ldots, r^{(t)}\}$ denote the sequence of relative growth rates. If three consecutive steps satisfy $r^{(t)} > r^{(t-1)}$, the neighborhood is considered to have loosened, and expansion halts. The adaptive upper radius of x_i is then defined as the average distance to its expanded neighborhood:

$$r_i^{\Uparrow} = \frac{1}{|\mathcal{S}_i|} \sum_{x_j \in \mathcal{S}_i} d(x_i, x_j). \tag{11}$$

The adaptive lower radius $r_i^{\Downarrow}$ is computed analogously for the lower granule $IC_B^{\Downarrow}(x_i)$.

This adaptive radius mechanism improves local structure awareness in granule construction, enabling precise modeling of both dense and sparse regions. It avoids the rigidity of fixed thresholds and enhances the granules' capability to reflect complex spatial patterns in real-world datasets.

2) Definition of R-Upper and R-Lower Cone Granules: Building upon the adaptive radius estimation described in Section III-B1, we construct localized upper and lower conical information granules for each object within their corresponding adaptive neighborhoods. Let $\Gamma^R_{EC^{\Uparrow}_B(x)}$ and $\Gamma^R_{IC^{\Downarrow}_B(x)}$ denote the R-upper and R-lower granules of object x, respectively. These granules capture local dominance semantics from complementary perspectives and are essential for accurately modeling object-level structural associations.

Definition 3.3: Let $S = \{U, A, f, V\}$ be an IS and $x_i \in U$. Given the upper conical granule $EC^{\Uparrow}_B(x)$, the R-upper conical granule (R-UG) of x is defined as:

$$\Gamma^R_{EC^{\Uparrow}_B(x)} = \{x_i \in EC^{\Uparrow}_B(x) \mid d(x, x_i) \leq r^{\Uparrow}_i\}. \tag{12}$$

Similarly, given the lower conical granule $IC^{\Downarrow}_B(x)$, the R-lower conical granule (R-LG) is defined as:

$$\Gamma^R_{IC^{\Downarrow}_B(x)} = \{x_i \in IC^{\Downarrow}_B(x) \mid d(x, x_i) \leq r^{\Downarrow}_i\}. \tag{13}$$

Once the R-upper and R-lower granules are computed for all objects, we construct the corresponding granule relation matrices, $EM^{\Uparrow}_R \in \mathbb{R}^{|U|\times|U|}$ and $LM^{\Downarrow}_R \in \mathbb{R}^{|U|\times|U|}$, to encode pairwise neighborhood dependencies:

$$EM^{\Uparrow}_R(x_i, x_j) = \begin{cases} 1, & \text{if } x_j \in \Gamma^R_{EC^{\Uparrow}_B(x_i)}; \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

$$LM^{\Downarrow}_R(x_i, x_j) = \begin{cases} 1, & \text{if } x_j \in \Gamma^R_{IC^{\Downarrow}_B(x_i)}; \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

In these matrices, the i-th row indicates the set of samples covered by the i-th object's granule, while the j-th column indicates the granules that include the j-th sample. This dual-view representation enables structured modeling of local neighborhood dependencies from both upper and lower dominance perspectives.

By introducing adaptive neighborhood constraints, the R-upper and R-lower granules extend the original conical granule model without altering its underlying dominance semantics. This refinement improves the representativeness of local structures, enhances robustness to noise and heterogeneous densities, and strengthens the expressive and discriminative capability of the granular clustering model, while remaining compatible with existing conical granule-based frameworks.

Fig. 1 illustrates the conical granule structure with adaptive radius constraints, showing the upper and lower bounds of the granular cone.

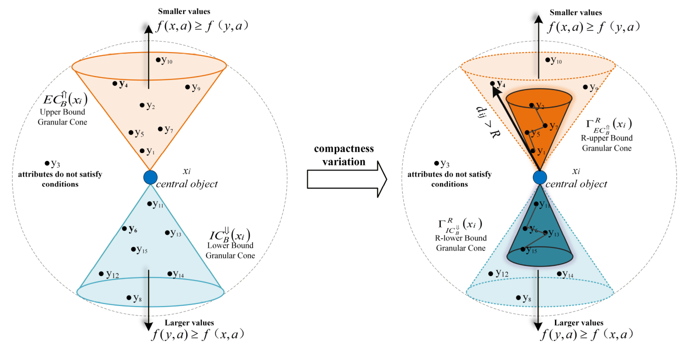


Fig. 1. Schematic of Conical Granular Structure with Radius Constraints.

C. Cone Granule-Based Similarity Measure

Accurate similarity assessment between data objects is crucial for effective clustering. Traditional metrics, such as Euclidean distance, often fail to capture intrinsic structural relationships in complex or non-Euclidean spaces. The previously defined R-upper and R-lower cone granules encode local data distribution patterns from dual semantic perspectives. Leveraging these granules, we propose a similarity measure based on cone granule co-coverage patterns, which emphasizes structural consistency and local semantics, ensuring that identified neighbors are both spatially proximate and semantically coherent.

D. Notation Clarification and Granule Matrices

To explicitly distinguish the contributions from two perspectives, we define coverage strength matrices for the upper and lower conical granules.

Definition 3.4: Let $S = (U, A, f, V)$ be an IS, $EM^{\Uparrow}_R$, $LM^{\Downarrow}_R \in \mathbb{R}^{|U|\times|U|}$. For each object $x_i \in U$, let $\Gamma^R_{EC^{\Uparrow}_B}(x_i)$ and $\Gamma^R_{IC^{\Downarrow}_B}(x_i)$ denote its R-upper and R-lower conical granules, respectively.

The upper coverage strength matrix $\mathbf{CA}^{\Uparrow} \in \mathbb{R}^{|U|\times|U|}$ is defined as:

$$\mathbf{CA}^{\Uparrow}(x_i, x_j) = \begin{cases} \frac{EM^{\Uparrow}_R(x_i, x_j)}{\sqrt{|\Gamma^R_{EC^{\Uparrow}_B}(x_i)|}}, & \text{if } x_j \in \Gamma^R_{EC^{\Uparrow}_B}(x_i), \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

Similarly, the lower coverage strength matrix $\mathbf{CA}^{\Downarrow} \in \mathbb{R}^{|U|\times|U|}$ is defined as:

$$\mathbf{CA}^{\Downarrow}(x_i, x_j) = \begin{cases} \frac{LM^{\Downarrow}_R(x_i, x_j)}{\sqrt{|\Gamma^R_{IC^{\Downarrow}_B}(x_i)|}}, & \text{if } x_j \in \Gamma^R_{IC^{\Downarrow}_B}(x_i), \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

The values $\mathbf{CA}^{\Uparrow}(x_i, x_j)$ and $\mathbf{CA}^{\Downarrow}(x_i, x_j)$ quantify the structural association strength between objects x_i and x_j from the upper and lower conical perspectives, respectively.

As defined above, cone granules that cover more samples are less discriminative due to their generality, while those covering fewer samples contribute more detailed and specific structural information. To enhance discriminability, we introduce a weighting strategy that adjusts the influence of each granule based on its size and semantic direction.

- *R-upper granules (R-UG):* larger granules imply broader coverage and are assigned higher weights:

$$w^+_k = \log\left(\frac{\left|\Gamma^R_{EC^{\Uparrow}_B(x_k)}\right|}{|U|} + 1\right) \tag{18}$$

- *R-lower granules (R-LG):* smaller granules are considered more specific and hence receive higher weights:

$$w_k^- = \log\left(\frac{|U|}{\left|\Gamma^R_{IC_B^{\Downarrow}(x_k)}\right|}\right) \tag{19}$$

The weights are incorporated into the coverage-strength matrices to yield the weighted coverage matrices $\mathbf{G}^{\Uparrow}, \mathbf{G}^{\Downarrow} \in \mathbb{R}^{|U|\times|U|}$:

$$\mathbf{G}^{\Uparrow} = \mathbf{CA}^{\Uparrow} \cdot \mathrm{diag}(\mathbf{w}^+),$$
$$\mathbf{G}^{\Downarrow} = \mathbf{CA}^{\Downarrow} \cdot \mathrm{diag}(\mathbf{w}^-). \tag{20}$$

Next, we derive the upper co-occurrence matrix $\mathbf{C}_{\text{adv}}$ and the lower counterpart $\mathbf{C}_{\text{dis}}$, which count how frequently two samples co-occur within the same cone granule:

$$\mathbf{C}_{\text{adv}} = (\mathbf{G}^{\Uparrow})^\top \mathbf{G}^{\Uparrow},$$
$$\mathbf{C}_{\text{dis}} = (\mathbf{G}^{\Downarrow})^\top \mathbf{G}^{\Downarrow}. \tag{21}$$

In these matrices, each entry $\mathbf{C}_{\text{adv}}(i,j)$ and $\mathbf{C}_{\text{dis}}(i,j)$ reflects the joint structural similarity between x_i and x_j as measured by shared coverage patterns.

The upper-view and lower-view similarities characterize complementary structural relations, yet each view alone may still be affected by noise or one-sided structural bias. As a result, high similarity inferred from a single perspective does not necessarily indicate reliable structural agreement between samples. To address this issue, we adopt a consistency-oriented integration principle, which emphasizes mutual support from both views when constructing the final similarity. Finally, after normalizing $\mathbf{C}_{\text{adv}}$ and $\mathbf{C}_{\text{dis}}$ to account for varying granule sizes and dataset scale, the resulting matrices serve as the upper-view similarity $\overline{\mathbf{C}}_{\text{adv}}$ and the lower-view similarity $\overline{\mathbf{C}}_{\text{dis}}$, which are fused element-wise to yield the final dual-perspective similarity measure:

$$S_f(x_i,x_j) = \overline{\mathbf{C}}_{\text{adv}}(x_i,x_j) \cdot \overline{\mathbf{C}}_{\text{dis}}(x_i,x_j) \tag{22}$$

This fusion mechanism emphasizes agreement across both semantic views. A high final similarity score is obtained only when both dominance (advantageous) and subordination (disadvantageous) perspectives align. This reduces noise sensitivity from either view and improves the robustness and discriminative power of the similarity assessment.

E. Clustering Process

To capture the neighborhood structures of data under multiple semantic perspectives, we construct adaptive R-upper and R-lower cone granules. A similarity measure is then designed by fusing co-coverage patterns of these granules, which enhances robustness and expressiveness in complex distributions. This section introduces the clustering model built upon the cone granule structure and similarity matrix, including center selection, assignment strategy, and optimization objectives.

1) Center Selection: The selection of cluster centers is crucial, as it determines the initial structure of clustering. Each center should be both representative of its local region and sufficiently distinct from others to avoid redundancy. To achieve this, we propose a center selection mechanism that integrates structural density and spatial dispersion using the adaptive cone granule structure.

Based on the previously construct $EM_R^{\Uparrow}$ and $LM_R^{\Downarrow}$, we construct the adjacency matrices of the upper and lower cone granules. These matrices are then concatenated to form a structural matrix $\mathbf{CS} \in \mathbb{R}^{N\times 2\,N}$:

$$\mathbf{CS} = \left[\, EM_R^{\Uparrow} \;\middle|\; LM_R^{\Downarrow} \right] \tag{23}$$

Both $EM_R^{\Uparrow}$ and $LM_R^{\Downarrow}$ are stored in Compressed Sparse Row (CSR) format.

The dual granules reflect complementary perspectives of subordination and dominance, forming the basis for centrality evaluation.

Definition 3.5: Let $S = \{U, A, f, V\}$ be an IS, and $x_i \in U$. Define:

- *Passive centrality:*

$$v^{\mathrm{e}}_{x_i} = \frac{1}{|U|}\sum_{j=1}^{|U|} EM_R^{\Uparrow}(x_j, x_i)$$

  Lower values indicate proximity to the structural center.
- *Active centrality:*

$$v^{\mathrm{l}}_{x_i} = \frac{1}{|U|}\sum_{j=1}^{|U|} LM_R^{\Downarrow}(x_j, x_i)$$

  Higher values imply stronger structural dominance.

Let the corresponding centrality vectors be:

$$\mathbf{v}^{\mathrm{e}} = \begin{bmatrix} v^{\mathrm{e}}_{x_1} \\ \vdots \\ v^{\mathrm{e}}_{x_N} \end{bmatrix} \qquad \mathbf{v}^{\mathrm{l}} = \begin{bmatrix} v^{\mathrm{l}}_{x_1} \\ \vdots \\ v^{\mathrm{l}}_{x_N} \end{bmatrix} \tag{24}$$

A compound centrality vector is defined as:

$$\mathbf{V} = \begin{bmatrix} 1 - \mathbf{v}^{\mathrm{e}} \\ \mathbf{v}^{\mathrm{l}} \end{bmatrix} \in \mathbb{R}^{2N} \tag{25}$$

Specifically, the passive centrality measures the frequency of sample control, reflecting the degree of proximity to the structural core; the active centrality measures the sample's control ability, reflecting its structural representativeness and dominance. These two aspects are semantically complementary, but they are not equivalent in terms of dimension and mechanism of action, and thus are not suitable for direct fusion using a single scalar. Based on this, we construct a composite vector to explicitly model the two types of centrality, thereby retaining their discriminative information without introducing parameters, and maintaining consistency in form with the fusion process of the $\mathbf{CS}$ matrix, enabling the selection of the core to consider both coreness and representativeness under structural constraints.

The centrality score of sample x_i is computed as:

$$\mathrm{Score}(x_i) = \sum_{j\in\mathcal{J}_i} \mathbf{CS}_{i,j} \cdot \mathbf{V}_j, \tag{26}$$

where $\mathcal{J}_i = \{j \mid \mathbf{CS}_{i,j} \neq 0\}$ indexes nonzero entries in row i of $\mathbf{CS}$.

Samples are ranked by score, and the top-ranked one is selected as the first cluster center and added to the center set $\mathcal{Z}$.

To ensure diversity among centers, we introduce a structure-based dissimilarity matrix $dis \in \mathbb{R}^{N\times N}$ using Jaccard distance:

Definition 3.6: Let $S = \{U, A, f, V\}$ be an IS and $x_i, x_j \in U$. Define:

$$dis(x_i, x_j) = 1 - \frac{|\mathbf{CS}_i \cap \mathbf{CS}_j|}{|\mathbf{CS}_i \cup \mathbf{CS}_j|}$$

where $\mathbf{CS}_i$ denotes the index set of nonzero entries in row i.

Subsequent centers are selected via a greedy strategy. At each step, choose the sample with the highest center selection criterion (CSC):

$$\text{CSC}(x_i) = \text{Score}(x_i) \cdot \min_{j\in\mathcal{Z}} (dis(x_i, x_j)) \tag{27}$$

Repeat until k centers are selected. This strategy ensures global structural awareness and local representativeness.

2) Assignment Process: This step assigns non-center samples to clusters based on semantic consistency from cone granule structures. A stepwise strategy is adopted to iteratively update memberships and centers, maximizing intra-cluster similarity and inter-cluster separation.

For each non-center sample x_i, we use the similarity matrix S_f to assess compatibility with existing clusters.

Definition 3.7: Let $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_k\}$ denote the set of clusters, where $\mathcal{C}_j$ represents the j-th cluster. Define:

$$\text{Sim}(x_i, CT_j) = \frac{1}{|CT_j|} \sum_{x_k\in CT_j} S_f(x_i, x_k)$$

Considering the influence of cluster centers, we define the final assignment score:

$$\text{Score}_j(x_i) = S_f(x_i, c_j) \cdot \text{Sim}(x_i, CT_j) \tag{28}$$

where $z_j \in \mathcal{Z}$ is the center of cluster j.

Each x_i is assigned to the cluster maximizing $\text{Score}_j(x_i)$. Upon assignment change, the affected cluster is updated. New centers are selected by maximizing similarity within each cluster:

$$c_j^{\text{new}} = \arg\max_{x_l\in CT_j} \sum_{x_k\in CT_j} S_f(x_l, x_k) \tag{29}$$

This process repeats for a fixed number of iterations, progressively refining clusters and centers to enhance convergence toward structural similarity. The approach mitigates early assignment errors while improving compactness and center representativeness.

Based on the preceding analysis, this section presents the step-by-step procedure of the proposed algorithm–Adaptive Clustering via Bidirectional Granules (ADGC):

1) *Construction of Bidirectional Cone Granule Structures:* The data is first initialized and preprocessed. Using a structural consistency threshold θ, we define the coverage scope for constructing upper and lower cone granules. To enhance adaptivity, the cone radius for each sample is estimated based on local structural variation, forming adaptive R-upper and R-lower cone granules. These bidirectional granules effectively capture complex structural associations and enrich semantic neighborhood representation.
2) *Center Selection:* A composite centrality indicator is derived by combining the control ability (from upper granules) and the frequency of being controlled (from lower granules), resulting in a unified centrality vector. A greedy selection strategy is employed to iteratively choose cluster centers by maximizing both centrality and structural diversity. This ensures that selected centers are representative and well-dispersed.
3) *Cone-Based Similarity Measurement:* The R-upper and R-lower granules jointly model both Euclidean and non-Euclidean relations. We first define a coverage strength metric to quantify intra-granule discrimination. Then, granule-specific weighting strategies are applied to construct a weighted coverage strength matrix. A co-occurrence matrix is derived to measure shared coverage between sample pairs. Finally, two similarity matrices (from advantageous and disadvantageous perspectives) are fused via element-wise multiplication to obtain a robust and discriminative similarity measure.
4) *Iterative Clustering Assignment:* Each sample is assigned to a cluster based on a combined score incorporating its similarity to the cluster center and its consistency with current cluster members. If the assignment changes, the affected cluster's center is updated immediately to maintain internal cohesion. This process iterates until convergence, yielding a structurally stable clustering configuration.

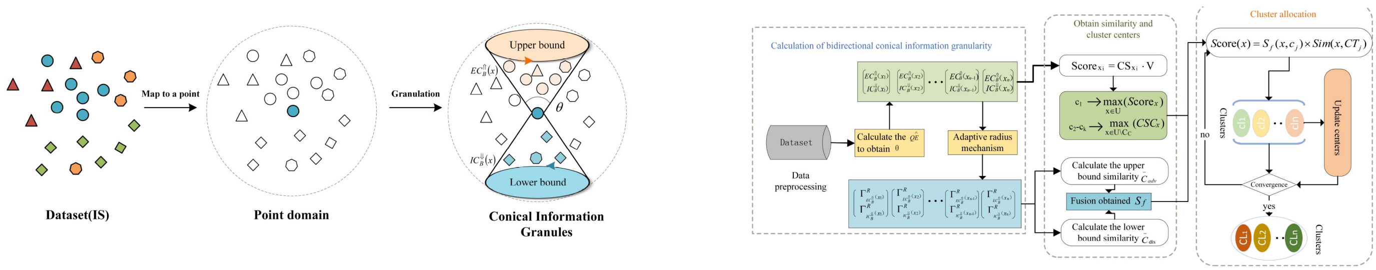
The complete workflow of the ADGC algorithm is illustrated in Fig. 2, and the detailed procedures are provided in Algorithms 1 to 2.

F. Complexity Analysis

The computational complexity of ADGC consists of two stages: cone-granule structure construction and structure-similarity-based clustering. Let n be the number of samples, d the feature dimensionality, k the number of cluster centers, and r the average number of non-zero elements per row in the structural matrix.

In Algorithm 1, quantile entropy computation requires $O(dn\log n)$ time, and evaluating attribute dominance relations for all sample pairs costs $O(n^2\ d)$.Radius estimation and cone-granule pruning take $O(n^2)$ time in the worst case.Thus, the time complexity of the first stage is $O(n^2\ d)$.

In Algorithm 2, selecting initial cluster centers costs $O(n^2)$,while constructing and fusing the structural similarity matrix requires $O(n^2 r)$ time.Sample assignment and center updating cost $O(n^2)$ per iteration, with $O(n^2/k)$ time for center recomputation in the worst case. Given a maximum of t iterations, the second stage has a time complexity of $O\big(n^2 r + t(n^2 + n^2/k)\big)$.Since the cone-granule structure grows sublinearly, r can be approximated as $O(n^{1/2})$,and when $k \ll n$, the complexity of Algorithm 2 reduces to $O(tn^2 + n^{5/2})$.



(a) (b)

Fig. 2. Overview of the proposed ADGC method. (a) Schematic illustration of granularity-based conical information granule construction. (b) Flowchart of the ADGC clustering algorithm.

Overall, the time complexity of ADGC is $O\big(n^2(d+t)+n^{5/2}\big)$,which can be approximated as $O(n^{5/2})$ when t is treated as a constant.

F.

a) Convergence Analysis: ADGC adopts an alternating optimization strategy. With the cluster center set C fixed, sample assignments are updated by maximizing $\mathrm{Score}_j(x_i)$ for each sample. With the partition t fixed, the cluster centers are updated according to Eq.(29) by selecting, within each cluster, the most representative sample that maximizes the sum of intra-cluster similarities. Both update steps perform maximization over finite candidate sets, which guarantees that a global structural consistency measure is monotonically non-decreasing throughout the iterations. Since the similarity matrix S_f is bounded and the joint state space of cluster assignments and center selections is finite, the objective value cannot increase indefinitely. Therefore, ADGC is guaranteed to converge to a stable solution in a finite number of iterations.

IV. Experiments

This section evaluates the effectiveness and robustness of the proposed ADGC algorithm through comprehensive experiments on publicly available datasets. All datasets are sourced from the UCI repository (https://archive.ics.uci.edu/datasets) and other manually curated collections. Comparative evaluations are conducted against a range of classical and granular computing-based clustering methods.

A. Baseline Methods and Implementation Details

To ensure a comprehensive and fair evaluation, the baseline methods are selected according to three widely accepted criteria in clustering research: (1) classical representative clustering algorithms, (2) state-of-the-art subspace and scalable clustering approaches, and (3) recent granular computing-based clustering methods closely related to this work. Specifically, the selected baseline methods are summarized as follows:

1) Classical and partition-based methods: K-means++ [20], CLARANS [30], Interpretable Subspace Clustering (ISC) [31], and One-step Bipartite Graph Cut (OBCut) [32]. These methods represent widely used clustering paradigms, including centroid-based and graph-based clustering.

2) Granular computing-based methods: Granular Ball Clustering Technique (GBCT) [13], Granular Ball Density Peaks (GB-DP) [12], Granular Ball DBSCAN (GB-DBSCAN) [11], K-Advantage Granularity Hierarchical Clustering (K-DGHC) [7], Bidirectional Cone-Based Clustering Granule (CBCG) [29], Granular Ball K-means (GBk-means) [33], and Granular Ball Spectral Clustering based on POJG (GB-POJG) [15]. These methods are closely related to the proposed approach and reflect recent advances in granular computing-based clustering. For simplicity, K-DGHC, GB-DP, GB-DBSCAN, GBk-means, and GB-POJG are abbreviated as KDG, GBDP, GBDB, GBKM, and POSC, respectively.

Implementation details: CLARANS, KDG, GBDP, GBDB, GBCT, CBCG, GBKM, and ADGC are implemented in Python, while K-means++ is utilized via the `scikit-learn` library. The implementations of ISC, OBCut, GBk-means, and GB-POJG are based on their original MATLAB codes to ensure consistency with the reported results. To guarantee fairness, all baseline methods are implemented using publicly available source codes or original authors' implementations. The parameters are carefully tuned according to recommended settings in the literature or determined via cross-validation when applicable, thereby avoiding potential bias in performance comparison.

Remark: We acknowledge that classical methods such as spectral clustering and DBSCAN are important baselines. In this study, we prioritize granular variants (e.g., GB-DBSCAN and GB-POJG) because they are more compatible with the granular computing framework adopted in this work, enabling structurally consistent comparisons. Incorporating additional classical baselines will be considered in future work to further strengthen the evaluation.

Table I summarizes the granulation mechanisms of the compared methods, highlighting the differences in granule geometry (e.g., granular balls vs. conical granules) and structural modeling (e.g., unidirectional vs. bidirectional relations).

The datasets used span a wide range of domains and complexity levels. Table II summarizes their characteristics, including sample size, dimensionality, and the number of clusters. To ensure fairness and reproducibility, we apply the following preprocessing steps:

Algorithm 1: Adaptive Bidirectional conical granule Construction with Dynamic Threshold.

Input: An IS: $T = \{U, A, f, V\}$
Output: R-upper and R-lower bound conical granule matrix $EM_R^{\Uparrow}, LM_R^{\Downarrow}$

$N \leftarrow |U|, DB^{Uparrow} \leftarrow \mathbf{0}_{N\times N}, \quad IDB^{\Downarrow} \leftarrow \mathbf{0}_{N\times N}$
for $d \leftarrow 1$ **to** $|A|$ **do**
  calculates $QE(d)$ according to Defnation 3.1.//Normalized to $\hat{QE}$
calculates P according to Eq.(7)
θ is calculated by the Eq.(8)
for $i \leftarrow 1$ **to** N **do**
  for $j \leftarrow 1$ **to** N **do**
    $count_{ij}^{+} \leftarrow \sum_{d=1}^{D}[x_i^{(d)} \geq x_j^{(d)}]$;
    $count_{ij}^{-} \leftarrow \sum_{d=1}^{D}[x_i^{(d)} \leq x_j^{(d)}]$;
    if $count_{ij}^{+} \geq \theta \cdot D$ **then**
      $DB^{\Uparrow}[i,j] \leftarrow 1$
    if $count_{ij}^{-} \geq \theta \cdot D$ **then**
      $IDB^{\Downarrow}[i,j] \leftarrow 1$
$EM_R^{\Uparrow} \leftarrow DB.copy(), \quad LM_R^{\Downarrow} \leftarrow IDB.copy()$
for $i \leftarrow 1$ **to** N **do**
  $M_i^{+} \leftarrow \{j \mid DB^{\Uparrow}[i,j] = 1\}, \quad M_i^{-} \leftarrow \{j \mid IDB^{\Downarrow}[i,j] = 1\}$
  calculates $r_i^{\Uparrow}$ according to Eq.(9)-(11)
  calculates $r_i^{\Downarrow}$ according to Eq.(9)-(11)
  foreach $j \in M_i^{+}$ **do**
    if $Dist[i,j] > r_i^{\Uparrow}$ **then**
      $EM^{\Uparrow}[i,j] \leftarrow 0$
  foreach $j \in M_i^{-}$ **do**
    if $Dist[i,j] > r_i^{\Downarrow}$ **then**
      $LM^{\Downarrow}[i,j] \leftarrow 0$
return $EM_R^{\Uparrow}, LM_R^{\Downarrow}$

- Remove samples with missing values;
- Apply min-max normalization to each attribute to mitigate scale effects;
- Use fixed seeds for stochastic algorithms to ensure consistency.

Clustering performance is evaluated using four widely adopted metrics:

- *Accuracy (ACC) [34]:* Measures the proportion of correctly clustered samples under optimal label alignment;
- *Normalized Mutual Information (NMI) [35]:* Quantifies the shared information between predicted and true cluster assignments;
- *Rand Index (RI) [36]:* Reflects the pairwise agreement between clustering and ground truth;
- *Silhouette Coefficient (SC) [37]:* Evaluates the compactness and separability of clusters.

Algorithm 2: Clustering Based on the Bidirectional Conical Granule Method.

Input: Number of clusters k; maximum iterations num_iter; bidirectional conical granule matrices $(EM_R^{\Uparrow}, LM_R^{\Downarrow})$ obtained by Algorithm 1
Output: $Label$: final clustering result

Initialize $Label \leftarrow$ zeros$(|U|, 1)$, $\mathcal{Z} \leftarrow \emptyset$, $rem \leftarrow U$;
$CS \leftarrow$ hstack(csr$(EM_R^{\Uparrow})$, csr$(LM_R^{\Downarrow})$);
Compute v_i^e, v_i^l $(i = 1, \ldots, N)$ by Definition 3.5, construct $\mathbf{V}$ by Eq. (25).
$score \leftarrow CS \cdot \mathbf{V}$
Select the maximum score index to add to $\mathcal{Z}$.
$rem \leftarrow rem \cup (U \setminus \mathcal{Z})$
Compute $dist[i,j]$ $(i, j = 1, \ldots, N)$ according to Definition 3.6;
for $t \leftarrow 2$ **to** k **do**
  foreach $j \in rem$ **do**
    $min\_dist(j) \leftarrow \max_{s\in\mathcal{Z}} dist[s,j]$;
  $CSC \leftarrow score(rem) \times min\_dist(rem)$, $z^{*} \leftarrow rem[\arg\max(CSC)]$;
  $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{z^{*}\}$, $rem \leftarrow rem \setminus \{z^{*}\}$;
Compute $\mathbf{CA}^{\Uparrow}, \mathbf{CA}^{\Downarrow}$ by Definition 3.4, and compute w_k^{+}, w_k^{-} by Eq. (18);
Compute S_f according to Eq. (21)–(22) and assign initial labels for centers in $\mathcal{Z}$;
for $t \leftarrow 1$ **to** num_iter **do**
  for $i \leftarrow 1$ **to** N **do**
    if $i \notin \mathcal{Z}$ **then**
      for $j \leftarrow 1$ **to** k **do**
        $\mathrm{Sim}(x_i, C_j) \leftarrow \frac{1}{|C_j|}\sum_{x_\ell\in C_j} S_f(x_i, x_\ell)$;
        $\mathrm{Score}_j(x_i) \leftarrow S_f(x_i, z_j) \cdot \mathrm{Sim}(x_i, C_j)$;
      $Label[i] \leftarrow \arg\max_j \mathrm{Score}_j(x_i)$;
  for $j \leftarrow 1$ **to** k **do**
    $C_j \leftarrow \{i \mid Label[i] = j\}$; **if** $C_j \neq \emptyset$ **then**
      compute z_j^{new} according to Eq. (29) and update $\mathcal{Z}$;
return $Label$;

B. Performance Comparison and Analysis

In this section, we evaluate the proposed ADGC algorithm against eleven representative clustering methods on 20 benchmark datasets listed in Table II. The baseline methods include K-means++ [20], CLARANS [30], ISC [31], OBCut [32], GBCT [13], GBDB [11], GBDP [12], CBCG [29], KDG [7], GBKM [33], and POSC [15]. For a fair comparison, all methods follow the parameter ranges suggested in their original literature, and grid search is applied to select the best configurations under identical evaluation criteria. All experiments are repeated 10 times with different random seeds, and average results are reported.

Table III lists the parameter settings of all methods, where k denotes the number of clusters, *ratio* controls the granularity

TABLE I
GRANULATION MECHANISMS OF RECENT GRANULAR CLUSTERING METHODS

Method	Granule type	Key mechanism
K-DGHC	Unidirectional conical granules	Hierarchical clustering via k-neighborhood dominance relations.
GB-DP	Granular balls	Density peak estimation over spherical local regions.
GB-DBSCAN	Granular balls	Granule-level density expansion, extending DBSCAN.
CBCG	Bidirectional conical granules	Ordinal dominance with mutual structural dependencies.
GBCT	Granular balls	Clustering via inter-ball structural relationships.
GBKM	Granular balls	Granular-ball enhanced prototype clustering.
POSC	Granular balls	Justifiable granularity balls with spectral clustering.
ADGC (Ours)	Bidirectional conical granules	Entropy-driven adaptive thresholding & growth-rate radius truncation.

The bold entry (i.e., "ADGC (Ours)") is used solely to highlight the proposed method and distinguish it from the existing approaches; it does not indicate a numerical superiority.

TABLE II
SUMMARY OF EXPERIMENTAL DATASETS

Dataset	Objects	Attributes	Clusters
hepatitis	112	18	2
track	163	7	2
Qualitative Bankruptcy (QB)	250	6	2
blobs	300	2	3
dermatology	358	34	6
Early Stage Diabetes (ED)	520	15	2
Statlog Heart (SH)	270	13	2
User Knowledge (UK)	258	5	4
Diabetes Data (DD)	520	16	2
pima	768	8	2
Raisin	900	7	2
ionosphere	351	34	2
zoo	101	16	7
TCGA	839	23	2
overlap	1000	2	6
OBS Network (ON)	1060	18	3
titanic	2201	3	2
Churn	3150	13	2
ring	7400	20	2
Electrical Grid (EG)	10000	13	2

level, K represents the number of neighbors in the local neighborhood, λ and γ in ISC balance self-reconstruction and group sparsity, M is the number of learned anchors, α weights the max and mean intra-cluster distances, and γ, δ in POSC control specificity impact and granularity threshold.

Table IV–VII provides a detailed comparison of performance indicators between ADGC and seven different algorithms. The bold values in each row represent the clustering analysis algorithms with the best performance on the current dataset.

From Table IV, ADGC achieves the best ACC on the majority of datasets (16 out of 20) and obtains the highest average ACC among all compared methods, demonstrating its strong

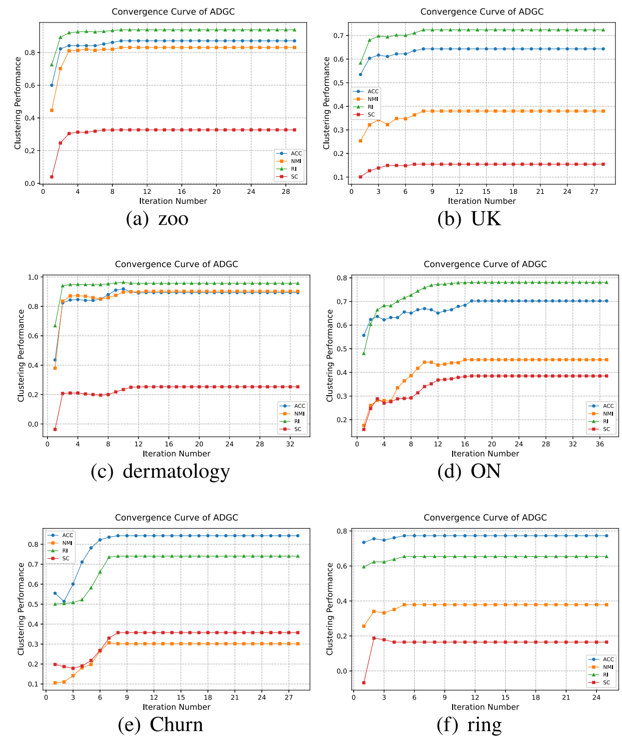


Fig. 3. The clustering convergence process of the ADGC algorithm on multiple typical datasets.

clustering capability and robustness across diverse data distributions.Notably, ADGC shows consistent improvements on datasets such as *Raisin* and *EG*. However, on the *TCGA* dataset, ISC achieves the best performance, indicating that certain methods may be more suitable for specific data characteristics. ADGC still maintains competitive results overall.These advantages can be attributed to the proposed dual-perspective conical granule modeling, which enhances the representation of local structural relationships, as well as the greedy center selection and adaptive iterative assignment strategies that further improve clustering stability and accuracy.

In Table V, ADGC achieves the best NMI on most datasets and yields the highest average NMI among all compared methods, indicating that it can better recover the underlying category structure and maintain high consistency with the ground-truth labels. This advantage is mainly related to its dual-perspective conical granule modeling strategy, which captures local structural information more accurately and improves the alignment between clustering results and true label distributions. It is worth noting that, on the *track* dataset, ADGC achieves the best ACC but not the best NMI. This suggests that although the predicted partition matches the labels well in terms of classification accuracy, its information consistency with the ground-truth partition is slightly weaker. Similar differences between ACC and NMI may occur when class boundaries are ambiguous or when the cluster structure does not perfectly coincide with the semantic label partition.

From Table VI, ADGC achieves the highest RI on the majority of datasets and the best average RI among all competing methods, demonstrating its strong ability to preserve pairwise

TABLE III
PARAMETER SETTINGS FOR ALGORITHMS ON DATASETS

DATASETS	K-means++	CLARANS	ISC	OBCut	GBDP	GBDB	KDG	CBCG	GBCT	GBKM	POSC	ADGC
hepatitis	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.04	k=2,K=52	k=2,K=7	k=2	$\alpha=0.7$, k=2	$\gamma=2$, $\delta=0.8$	k=2
track	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.05	k=2,K=77	k=2,K=12	k=2	$\alpha=0.7$, k=2	$\gamma=2$, $\delta=0.4$	k=2
QB	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.99	k=2,K=157	k=2,K=7	k=2	$\alpha=0.7$, k=2	$\gamma=1$, $\delta=0.1$	k=2
blobs	k=3	k=3	$\lambda=100, \gamma=0.1$	M=6	k=3	ratio=0.74	k=3,K=102	k=3,K=32	k=3	$\alpha=0.5$, k=3	$\gamma=1$, $\delta=0.1$	k=3
dermatology	k=6	k=6	$\lambda=100, \gamma=0.1$	M=12	k=6	ratio=0.26	k=6,K=51	k=6,K=12	k=6	$\alpha=0.6$,k=6	$\gamma=4$, $\delta=0.9$	k=6
ED	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.85	k=2,K=50	k=2,K=22	k=2	$\alpha=0.8$, k=2	$\gamma=1$, $\delta=0.1$	k=2
SH	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.03	k=2,K=112	k=2,K=67	k=2	$\alpha=0.5$, k=2	$\gamma=3$, $\delta=0.3$	k=2
UK	k=4	k=4	$\lambda=100, \gamma=0.1$	M=8	k=4	ratio=0.28	k=4,K=44	k=4,K=12	k=4	$\alpha=0.8$,k=4	$\gamma=10$, $\delta=0.4$	k=4
DD	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.85	k=2,K=402	k=2,K=22	k=2	$\alpha=0.5$,k=2	$\gamma=1$, $\delta=0.1$	k=2
pima	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.04	k=2,K=18	k=2,K=31	k=2	$\alpha=0.5$, k=2	$\gamma=1$, $\delta=0.7$	k=2
Raisin	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.02	k=2,K=172	k=2,K=67	k=2	$\alpha=0.7$, k=2	$\gamma=2$, $\delta=0.8$	k=2
ionosphere	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.07	k=2,K=109	k=2,K=24	k=2	$\alpha=0.4$,k=2	$\gamma=1$, $\delta=0.1$	k=2
zoo	k=7	k=7	$\lambda=100, \gamma=0.1$	M=14	k=7	ratio=0.68	k=7,K=22	k=7,K=11	k=7	$\alpha=0.3$,k=7	$\gamma=1$, $\delta=1$	k=7
TCGA	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.01	k=2,K=39	k=2,K=67	k=2	$\alpha=0.7$,k=2	$\gamma=1$, $\delta=0.75$	k=2
overlap	k=6	k=6	$\lambda=100, \gamma=0.1$	M=12	k=6	ratio=0.35	k=6,K=532	k=6,K=232	k=6	$\alpha=0.7$,k=6	$\gamma=2$, $\delta=0.95$	k=6
ON	k=3	k=3	$\lambda=100, \gamma=0.1$	M=6	k=3	ratio=0.73	k=3,K=207	k=3,K=52	k=3	$\alpha=0.5$,k=3	$\gamma=1$, $\delta=0.7$	k=3
titanic	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.10	k=2,K=21	k=2,K=32	k=2	$\alpha=0.7$,k=2	$\gamma=1$, $\delta=0.1$	k=2
Churn	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.05	k=2,K=30	k=2,K=46	k=2	$\alpha=0.6$,k=2	$\gamma=5$, $\delta=0.9$	k=2
ring	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.21	k=2,K=92	k=2,K=13	k=2	$\alpha=0.6$,k=2	$\gamma=5$, $\delta=0.9$	k=2
EG	k=2	k=2	$\lambda=100, \gamma=0.1$	M=4	k=2	ratio=0.04	k=2,K=32	k=2,K=57	k=2	$\alpha=0.4$,k=2	$\gamma=1$, $\delta=0.9$	k=2

TABLE IV
ACC COMPARISON OF 20 DATASETS

	Prototype-based				Granular computing-based							Proposed
DATASETS	K-means++	CLARANS	ISC	OBCut	GBDP	GBDB	KDG	CBCG	GBCT	GBKM	POSC	ADGC
hepatitis	0.6518	0.5893	0.5178	0.6607	0.5535	0.5804	0.5804	0.6696	0.6607	0.6244	0.6339	**0.7143**
track	0.7975	0.7791	0.5582	0.7484	0.5705	0.5398	0.7177	0.7662	0.7669	0.7922	0.5399	**0.8221**
QB	0.9800	0.9560	0.712	0.972	0.8600	0.9720	0.9840	0.8720	0.6440	0.974	0.98	**0.9920**
blobs	0.9633	0.9400	0.5733	0.9666	0.3600	0.9633	0.6200	0.96	0.9620	0.5105	0.96	**0.9667**
dermatology	0.7898	0.7458	0.4776	0.6536	0.2849	0.8352	0.7849	0.8496	0.7179	0.4682	0.3994	**0.8911**
ED	0.7462	0.7019	0.7788	0.6692	0.6173	0.7404	0.7903	0.8000	0.6212	0.7063	0.6154	**0.8135**
SH	0.5296	0.6963	0.6185	0.7296	0.5074	0.6296	0.7740	0.8074	0.7926	0.7107	0.6444	**0.8333**
UK	0.5736	0.5388	0.3255	0.4806	0.3217	0.5814	0.4689	0.6279	0.3721	0.4209	0.5736	**0.6434**
DD	0.7615	0.7019	0.7403	0.6865	0.6500	0.6654	0.6385	0.8096	0.7500	0.8269	0.6154	**0.8212**
pima	0.6680	0.6289	0.5442	0.6145	0.6197	0.6732	0.6562	**0.7070**	0.6497	0.5951	0.6602	0.6979
Raisin	0.7633	0.7956	0.7533	0.7411	0.6433	0.6866	0.8100	0.7355	0.8167	0.6589	0.6233	**0.8444**
ionosphere	0.7094	0.6952	0.5669	0.7122	0.5527	0.6980	0.6809	0.7009	0.6296	0.7749	0.6439	**0.7849**
zoo	0.6436	0.6535	0.4356	0.5742	0.2574	0.7425	0.6436	0.6535	0.5941	0.7030	0.8218	**0.8712**
TCGA	0.6698	0.6937	**0.8889**	0.8462	0.6448	0.5830	0.7151	0.8439	0.6222	0.7330	0.7342	0.8546
overlap	0.7120	0.7730	0.408	0.552	0.2000	0.816	0.4980	0.5150	0.7890	0.5430	0.522	**0.8300**
ON	0.6840	0.6311	0.5547	0.6556	0.4103	0.6509	0.5660	0.6226	**0.7264**	0.5142	0.4547	0.7028
titanic	0.7506	0.5334	0.5370	0.6819	0.6033	0.7051	0.6819	0.7742	0.6774	0.6792	0.6792	**0.7753**
Churn	0.7688	0.8295	0.5234	0.8295	0.7168	0.6994	0.8429	0.8419	0.8384	0.6905	0.6006	**0.8467**
ring	0.7507	0.5048	0.5894	0.7582	0.5108	0.5630	0.5049	0.5997	0.5245	0.6385	0.5919	**0.7777**
EG	0.6062	0.5765	0.6886	0.8018	0.6075	0.6410	0.6378	0.7873	0.6295	0.6385	0.6945	**0.8309**
AVG ACC	0.7260	0.6982	0.5896	0.7167	0.5246	0.6983	0.6798	0.7472	0.6892	0.6651	0.6443	**0.8157**

The bold values in each row represent the clustering analysis algorithms with the best performance on the current dataset.

relationships between samples. Although it is not always the top performer on every dataset, ADGC consistently maintains competitive performance, reflecting its robustness across diverse data distributions. This improvement can be attributed to the dual-perspective conical granule modeling, which captures complementary local structural information and enhances the discriminative power of anchor representations, leading to more consistent and stable clustering results.

As shown in Table VII, ADGC exhibits competitive SC performance on most datasets and achieves the highest average SC among all compared methods, indicating a favorable balance between intra-cluster compactness and inter-cluster separation. These results suggest that the bidirectional conical granule design helps ADGC capture structural characteristics of the data more effectively, thereby improving cluster cohesion and separability. It is also worth noting that, on datasets such as *overlap*, *track*, and *titanic*, ADGC does not achieve the best SC although its ACC remains very high. This is because SC mainly evaluates geometric compactness and separation, whereas ACC measures the consistency between clustering results and ground-truth labels. Therefore, on datasets with overlapping regions or ambiguous boundaries, a method may achieve high ACC even when its geometric separability is not optimal.

Overall, the ADGC algorithm demonstrates robust and distinctive clustering capabilities across multiple dimensions:

TABLE V
NMI COMPARISON OF 20 DATASETS

	Prototype-based				Granular computing-based							Proposed
DATASETS	K-means++	CLARANS	ISC	OBCut	GBDP	GBDB	KDG	CBCG	GBCT	GBKM	POSC	ADGC
hepatitis	0.0629	0.0121	0.0019	0.0698	0.0264	0.0001	0.0382	0.0134	0.0642	0.0599	0.0001	**0.1246**
track	**0.4068**	0.3415	0.0085	0.2066	0.0278	0.002	0.2819	0.2366	0.3563	0.4005	0.0001	0.3957
QB	0.8625	0.7438	0.2173	0.8222	0.4195	0.9047	0.8801	0.1578	0.3516	0.8939	0.8625	**0.9060**
blobs	0.8523	0.7964	0.2421	0.8555	0.0058	0.7921	0.3262	0.8516	0.8474	0.2996	0.8379	**0.8688**
dermatology	0.8446	0.7466	0.2623	0.7621	0.0373	0.8609	0.7335	0.8216	0.7881	0.4616	0.1332	**0.8980**
ED	0.2059	0.1297	0.2453	0.0644	0.0211	0.1766	0.0833	0.3360	0.0073	0.1961	0.0027	**0.4192**
SH	0.0005	0.1825	0.0398	0.1908	0.0024	0.1576	0.2384	0.2883	0.2614	0.1922	0.0001	**0.3537**
UK	0.3535	0.3697	0.0435	0.1930	0.0335	0.3475	0.1526	0.0809	0.0778	**0.4235**	0.2437	0.3799
DD	0.2279	0.1297	0.1747	0.0854	0.0616	0.1130	0.0026	0.3511	0.3299	0.3594	0.0027	**0.4316**
pima	0.0517	**0.1136**	0.0002	0.0286	0.0115	0.0733	0.0314	0.0450	0.0042	0.1086	0.0001	0.1114
Raisin	0.3310	0.3548	0.1960	0.2497	0.2137	0.1890	0.3577	0.2134	0.2684	0.2100	0.1602	**0.3783**
ionosphere	0.1299	0.0986	0.0421	0.1313	0.0001	0.1380	0.0424	0.0896	0.0006	0.2386	0.0087	**0.2258**
zoo	0.7451	0.6587	0.3752	0.6676	0.1560	0.7789	0.6981	0.0670	0.5782	0.6491	0.7218	**0.8251**
TCGA	0.1585	0.1760	**0.4935**	0.4044	0.3702	0.0244	0.1853	0.4017	0.1228	0.1667	0.1958	0.4288
overlap	0.7050	0.6607	0.2074	0.3951	0.0005	0.7055	0.3901	0.4739	0.7048	**0.7968**	0.3610	0.7054
ON	0.4004	0.3338	0.2169	0.3593	0.0070	0.3899	0.1924	0.4324	**0.4926**	0.1030	0.0001	0.4547
titanic	0.1365	0.0173	0.0064	0.0752	0.0001	0.0541	0.0016	0.1568	0.0160	0.0126	0.0108	**0.1675**
Churn	0.0225	0.1377	0.0005	0.1924	0.0003	0.2175	0.2568	0.2889	0.2167	0.0661	0.0001	**0.2965**
ring	0.2425	0.0001	0.0467	0.2245	0.0006	0.0001	0.0001	0.1395	0.0280	0.1149	0.0001	**0.3786**
EG	0.0368	0.0129	0.0770	0.2271	0.0001	0.0066	0.0003	0.1720	0.0010	0.0422	0.1227	**0.3755**
AVG NMI	0.3388	0.3008	0.1761	0.2966	0.0698	0.2966	0.2447	0.2809	0.2759	0.2849	0.2020	**0.4563**

The bold values in each row represent the clustering analysis algorithms with the best performance on the current dataset.

TABLE VI
RI COMPARISON OF 20 DATASETS

	Prototype-based				Granular computing-based							Proposed
DATASETS	K-means++	CLARANS	ISC	OBCut	GBDP	GBDB	KDG	CBCG	GBCT	GBKM	POSC	ADGC
hepatitis	0.5420	0.5116	0.4961	0.5476	0.5013	0.5085	0.5013	0.5058	0.5472	0.5320	0.5381	**0.5852**
track	0.6751	0.6537	0.5038	0.6212	0.0141	0.0008	0.1852	0.2616	0.6402	0.6777	0.5005	**0.7057**
QB	0.9606	0.9155	0.5882	0.9454	0.6692	0.9727	0.9684	0.5150	0.5396	0.9577	0.9606	**0.9762**
blobs	0.9530	0.9252	0.6538	0.9566	0.5567	0.9268	0.6649	0.9481	0.9490	0.5065	0.9487	**0.9566**
dermatology	0.9173	0.8776	0.7627	0.8750	0.6544	0.9224	0.9084	0.9210	0.8889	0.5574	0.7100	**0.9568**
ED	0.6205	0.5807	0.6549	0.5564	0.5266	0.6069	0.6703	0.6748	0.5284	0.6058	0.5248	**0.6959**
SH	0.4999	0.5755	0.5263	0.6040	0.4983	0.5801	0.6489	0.6833	0.6700	0.5999	0.5407	**0.7212**
UK	0.6984	0.7112	0.6162	0.6722	0.5707	0.6925	0.5542	0.3361	0.5939	0.7283	0.6706	**0.7241**
DD	0.6361	0.5807	0.6148	0.5688	0.5441	0.5451	0.5044	0.6911	0.6243	0.7132	0.5248	**0.7057**
pima	0.5558	0.5326	0.5033	0.5256	0.5281	0.5594	0.5516	0.5649	0.5443	0.5174	0.5515	**0.5778**
Raisin	0.6383	0.6743	0.6279	0.6158	0.5463	0.1557	0.3838	0.2212	0.6515	0.5500	0.5299	**0.7370**
ionosphere	0.5865	0.5750	0.5076	0.5889	0.5041	0.5772	0.5582	0.5684	0.5323	0.6502	0.5401	**0.6602**
zoo	0.8719	0.8400	0.7509	0.8416	0.6529	0.8877	0.8519	0.2689	0.8317	0.8228	0.8646	**0.9386**
TCGA	0.5572	0.5745	**0.8009**	0.7395	0.6606	0.5050	0.5921	0.7362	0.5293	0.6081	0.6092	0.7512
overlap	0.8843	0.8803	0.7670	0.8130	0.7064	0.8901	0.7375	0.7345	0.8821	0.5700	0.7421	**0.9028**
ON	0.7346	0.6811	0.6465	0.6980	0.4555	0.7442	0.5740	0.7207	0.7656	0.5123	0.5791	**0.7790**
titanic	0.6254	0.5020	0.5025	0.5660	0.5211	0.5692	0.5628	0.6502	0.5628	0.5641	0.5641	**0.6628**
Churn	0.6445	0.7171	0.5009	0.7171	0.5939	0.5886	0.7203	0.6567	0.7171	0.5724	0.3800	**0.7403**
ring	0.6256	0.4999	0.5159	0.6333	0.5002	0.4967	0.5000	0.5198	0.5011	0.5539	0.5182	**0.6542**
EG	0.5225	0.5117	0.5671	0.6793	0.5146	0.5397	0.5438	0.5944	0.5330	0.5383	0.5756	**0.7190**
AVG RI	0.6875	0.6660	0.6054	0.6883	0.5360	0.6135	0.6091	0.5886	0.6516	0.6158	0.6394	**0.7575**

The bold values in each row represent the clustering analysis algorithms with the best performance on the current dataset.

- *Bidirectional Conical Granule Modeling:* ADGC employs a dual-view modeling strategy that characterizes each sample from both upper and lower dominance perspectives. By adaptively extracting local features through forward upper coverage and backward lower constraints, it enhances both the discriminability and completeness of structural representations.
- *Multi-scale Structural Similarity:* The algorithm constructs a structural similarity matrix by integrating the coverage relationships of R-upper and R-lower granules. This dual-view measure captures fine-grained affinity between samples and supports more reliable similarity estimation.
- *Greedy Center Selection:* In the clustering phase, ADGC adopts a greedy strategy to identify initial cluster centers based on a composite centrality score that incorporates structural salience and dispersion. This reduces reliance on heuristic parameters and improves initialization robustness.
- *Adaptive Assignment Mechanism:* An iterative sample assignment strategy dynamically updates cluster

TABLE VII
SC Comparison of 20 Datasets

	Prototype-based				Granular computing-based							Proposed
DATASETS	K-means++	CLARANS	ISC	OBCut	GBDP	GBDB	KDG	CBCG	GBCT	GBKM	POSC	ADGC
hepatitis	0.1144	0.0381	0.2092	0.0637	–0.0121	0.0010	–0.0638	0.0023	0.1257	0.1529	0.4192	**0.2182**
track	–0.0142	–0.0255	0.0845	0.1170	0.057	0.2160	0.4108	0.3875	0.4180	0.3782	0.4761	**0.4562**
QB	0.3442	0.3238	0.1362	0.3431	0.1558	0.1211	0.3433	0.1754	0.1529	0.3344	0.3433	**0.3446**
blobs	**0.5589**	0.5277	0.0740	0.5421	0.0058	0.2252	0.2654	0.5421	0.5492	0.2455	0.5576	0.5506
dermatology	–0.0726	–0.0327	–0.0682	–0.0095	–0.1024	–0.0173	0.1495	0.1469	0.1611	0.0477	0.2165	**0.2250**
ED	0.1084	0.1365	0.0224	0.1369	–0.0033	–0.1842	0.1849	0.1941	0.0399	0.1765	0.1898	**0.1972**
SH	–0.0110	0.0153	**0.2180**	0.0098	–0.0019	–0.1580	0.1290	0.2035	0.1913	0.1472	0.0536	0.2056
UK	0.1074	0.1205	0.0286	0.1505	0.0001	0.0173	0.0640	0.0365	0.1029	0.1636	**0.1732**	0.1546
DD	0.0982	0.1363	0.0700	0.1333	0.0098	–0.5361	0.0261	0.1836	0.1745	0.1826	**0.2891**	0.1936
pima	0.0565	0.0012	0.0392	0.2957	–0.0094	0.2098	0.1858	0.2009	0.3153	0.0501	**0.3955**	0.2156
Raisin	0.3310	0.3548	0.2077	0.6131	–0.0833	–0.0094	0.3615	0.2685	0.3201	**0.5123**	0.3582	0.3747
ionosphere	0.2970	0.2886	0.0761	0.2960	0.093	0.2566	0.2101	0.2595	0.1697	0.1642	**0.4083**	0.3047
zoo	0.2077	0.0873	–0.0214	0.2132	–0.1658	0.1180	0.2439	–0.3294	0.1761	0.2382	0.3034	**0.3316**
TCGA	0.1505	0.1626	0.1492	0.2817	–0.1805	0.1468	0.1838	0.2032	0.1864	0.1794	0.1868	**0.2038**
overlap	0.4168	0.3420	–0.0199	0.1039	–0.0315	**0.4297**	0.1361	0.1460	0.3835	0.1015	–0.0015	0.3873
ON	0.2522	0.2156	–0.0381	0.2836	–0.1281	–0.2527	0.1486	0.3128	0.3427	0.2569	0.0488	**0.3704**
titanic	0.6010	0.3780	0.0124	0.5504	0.0203	–0.5692	0.5083	0.6202	0.4702	**0.6854**	**0.6854**	0.6390
Churn	0.0894	0.0885	0.0305	0.0894	–0.0044	–0.3789	0.2346	0.0200	0.3555	0.3064	0.3268	**0.3574**
ring	0.1246	0.0271	–0.0640	0.1077	0.0106	–0.3021	–0.0025	0.1544	0.1622	0.0927	0.0037	**0.1647**
EG	0.0313	0.0478	0.2068	**0.3465**	–0.0023	0.0101	–0.0415	0.0490	0.0079	0.1693	0.1477	0.0740
AVG SC	0.1896	0.1617	0.0677	0.2334	-0.0186	-0.0328	0.1839	0.1889	0.2403	0.2293	0.2790	**0.2984**

The bold values in each row represent the clustering analysis algorithms with the best performance on the current dataset.

memberships by exploiting intra-cluster structural consistency, enabling the model to adapt to complex distributions and maintain high clustering accuracy even in noisy or irregular data.

C. Effectiveness Analysis of the Center Selection Strategy

In most clustering algorithms, the initialization of cluster centers is a critical factor influencing the quality of final clustering results. This is particularly important for datasets with complex distributions or heterogeneous densities, where the representativeness and spatial diversity of initial centers significantly affect both clustering accuracy and stability. Inappropriate center initialization may lead to suboptimal solutions or incorrect data partitioning.

To address this issue, the proposed ADGC algorithm incorporates a structure-aware center selection mechanism that leverages bidirectional conical granule information and structural dispersion metrics to identify highly representative and well-separated initial centers. To systematically evaluate the effectiveness of this strategy, we design a controlled comparative experiment in which only the center selection module is replaced, while all other components of ADGC remain fixed.

The alternative center initialization strategies considered include:

- *RAN-GC:* Random selection from the dataset;
- *DP-GC:* Density peak-based initialization using cone-based distance;
- *C-KM:* KMeans++ initialization using cone-based similarity;
- *UMGC:* Upper-view-based center selection from $EC_B^{\Uparrow}$ only;
- *LMGC:* Lower-view-based center selection from $IC_B^{\Downarrow}$ only.

TABLE VIII
ACC Comparison of 20 Datasets

DATASETS	RAN-GC	C-KM	DP-GC	LMGC	UMGC	ADGC
hepatitis	0.6517	0.7054	0.6607	0.7054	0.7085	**0.7143**
track	0.5399	0.5337	0.8220	0.5699	**0.8251**	0.8221
QB	0.968	0.985	0.989	0.988	0.9780	**0.9920**
blobs	0.9600	0.9633	0.9567	0.9456	0.9663	**0.9667**
dermatology	0.7514	0.8156	0.8464	0.8631	0.7265	**0.8911**
ED	0.8058	0.8000	0.8057	0.8038	0.8077	**0.8135**
SH	0.8059	0.8022	0.7888	0.6592	0.8143	**0.8333**
UK	0.5891	0.5348	0.5698	0.5969	0.5853	**0.6434**
DD	0.7813	0.7700	0.6807	0.7935	0.8173	**0.8212**
pima	0.6796	0.6888	0.6875	0.6589	0.6914	**0.6979**
Raisin	0.8300	0.8400	0.8422	0.8434	0.8422	**0.8444**
ionosphere	0.7835	0.7493	0.7521	0.7721	0.7493	**0.7849**
zoo	0.5545	0.7822	0.8416	0.6248	0.8131	**0.8712**
TCGA	0.8415	0.8439	0.8510	0.8456	0.8379	**0.8546**
overlap	0.82240	0.8035	0.7080	0.8272	0.8186	**0.8300**
ON	0.5755	0.6509	0.6179	0.6887	0.6840	**0.7028**
titanic	0.5538	0.7532	0.5538	0.6769	0.7733	**0.7753**
Churn	0.5695	0.7987	0.7600	0.7905	0.8429	**0.8467**
ring	0.7579	0.7580	0.7549	0.7601	0.7548	**0.7777**
EG	0.7197	0.7280	0.7639	0.7830	0.8027	**0.8309**
AVG ACC	0.7271	0.7653	0.7626	0.7598	0.7920	**0.8157**

The bold value in each row denotes the highest ACC achieved among the evaluated initialization methods (including ADGC) on that dataset.

To ensure a fair comparison, all baseline strategies operate on the same cone-structured similarity matrices, thereby isolating the influence of center selection alone. Clustering accuracy (ACC) is used as the evaluation metric, and experiments are conducted on 20 datasets. The results are summarized in Table VIII.

Overall, ADGC achieves the best performance on most datasets and consistently maintains superior average accuracy across all benchmarks, demonstrating notable stability and adaptability. Compared to RAN-GC and C-KM, ADGC more effectively captures global structural distinctions and mitigates errors caused by poor local initialization. Relative to DP-GC, ADGC benefits from bidirectional granule modeling and structural centrality optimization, making it more resilient to irregular cluster shapes and complex distributions. Moreover, in contrast to single-view strategies such as UMGC and LMGC, which only consider either advantageous or disadvantageous neighborhood information, ADGC fuses information from both perspectives. This dual-view modeling enhances inter-cluster separability and yields a more balanced initialization scheme. In summary, the experimental results provide strong evidence for the effectiveness and robustness of the proposed structure-aware center selection strategy. These findings also highlight the critical role of semantically informed initialization in improving clustering outcomes under diverse data environments.

D. Convergence Analysis

To further validate the stability and optimization performance of the proposed ADGC algorithm during the clustering process, we conducted a series of convergence experiments. Specifically, we recorded the values of key clustering metrics–Accuracy (ACC), Normalized Mutual Information (NMI), Rand Index (RI), and Silhouette Coefficient (SC)–after each iteration and plotted the corresponding convergence curves. Fig. 3 presents the convergence behavior of ADGC on datasets of varying sizes. Convergence experiments were conducted across all 20 datasets, and representative cases–*zoo*, *U.K.*, *dermatology*, *ON*, *Churn*, and *ring*–are selected and displayed in ascending order of dataset size. From the plots, it is evident that all four metrics exhibit rapid improvement during the initial iterations, followed by a gradual stabilization. This indicates that ADGC typically converges within a limited number of iterations. In particular, ACC, RI, and NMI reach stable values within approximately 5 to 15 iterations on most datasets. Meanwhile, the SC metric shows a steadily increasing trend, reflecting continuous optimization in terms of intra-cluster compactness and inter-cluster separability. Furthermore, our experimental results suggest that convergence speed is not strongly correlated with dataset size, but is more sensitive to the number of clusters. Datasets with a larger number of clusters generally require more iterations to stabilize both the cluster centers and the sample assignments. This observation implies that the convergence behavior of ADGC is more influenced by the structural complexity of the clustering task than by the raw scale (i.e., sample size) of the data.

In addition, we incorporate a convergence criterion based on the proportion of label changes between successive iterations. Specifically, the iterative process is terminated early when the proportion of samples with changed cluster assignments falls below a predefined threshold, thereby avoiding unnecessary computations. In our implementation, this threshold is set to $\mathtt{tol} = 10^{-4}$.This early-stopping mechanism is designed to reduce runtime overhead without sacrificing clustering accuracy. Empirical results confirm that this strategy effectively improves algorithmic efficiency while maintaining consistent clustering performance. It enables the model to adaptively determine when convergence is achieved, particularly in scenarios where further iterations yield diminishing returns.

In summary, the proposed ADGC algorithm demonstrates favorable convergence properties and practical stability. It consistently reaches optimization objectives within a limited number of iterations and exhibits strong adaptability to diverse and complex data structures. These characteristics make ADGC well suited for efficient and scalable clustering applications.

V. Conclusion

This paper presents a novel clustering algorithm, ADGC, which integrates granular computing theory with dual-view structural modeling. The proposed method effectively captures the geometric and semantic structures of complex data by constructing bidirectional conical information granules under geometric and ordinal constraints.

Specifically, ADGC introduces a dynamic thresholding strategy using quantile entropy to determine a structural consistency threshold θ, followed by the construction of upper- and lower-bound conical granules to model dominance relationships. To refine local granularity, an adaptive radius estimation mechanism is designed based on the expansion trend of granule compactness. Furthermore, a dual-perspective similarity measure and a center-guided assignment strategy are employed to enhance clustering accuracy, stability, and interpretability.

ADGC achieves systematic improvements in multiple dimensions. First, dynamic granule construction reduces dependence on manual parameter tuning and enhances adaptability to varied data distributions. Second, the incorporation of structural centrality in center selection and allocation improves clustering robustness across complex scenarios. Extensive experiments on diverse benchmark datasets demonstrate the superior performance of ADGC in terms of accuracy, stability, and generalization.

Despite its advantages, several limitations remain. 1)In high-dimensional settings, the amplified sparsity of data progressively weakens ordinal-dominance relationships, thereby degrading the granulation performance of the conical-granulation framework that relies on such ordinal relations.The granule-based representation may be less effective in high-dimensional sparse spaces.2)In ADGC, a pairwise similarity relationship needs to be constructed among samples, with a time complexity of $O(N^{5/2})$. Although it can achieve effective global structure reasoning, it limits its scalability on large-scale datasets. In addition, the algorithm currently requires the number of clusters to be manually specified, which limits its full automation potential.

Future work will focus on two key directions. First, improving self-adaptivity by developing mechanisms that automatically estimate the number of clusters and adjust granule radii based on data distribution. Secondly, in large-scale and high-dimensional environments, we will improve the efficiency and scalability of the granulation strategy, such as by introducing methods like attribute selection, sparse granularity strategies, or approximate structural similarity calculations, thereby enhancing the applicability of the proposed model in complex real-world clustering tasks.

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